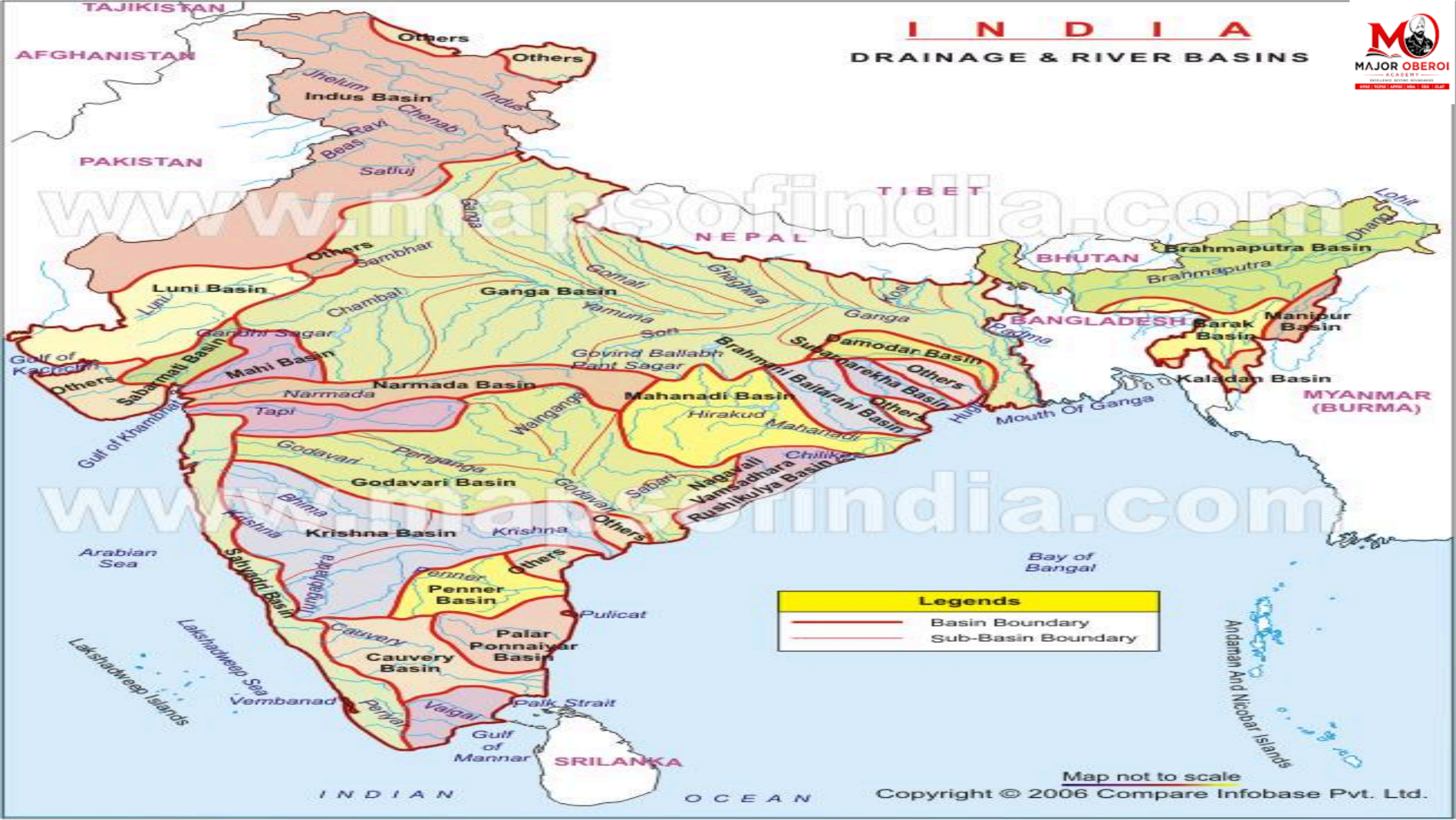


Drainage Systemof India

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The Indian Drainage System

- The Indian drainage system consists of a large number of small and big rivers.
- It is an outcome of:
- The evolutionary process of the three major physiographic units
- The nature and characteristics of precipitation.



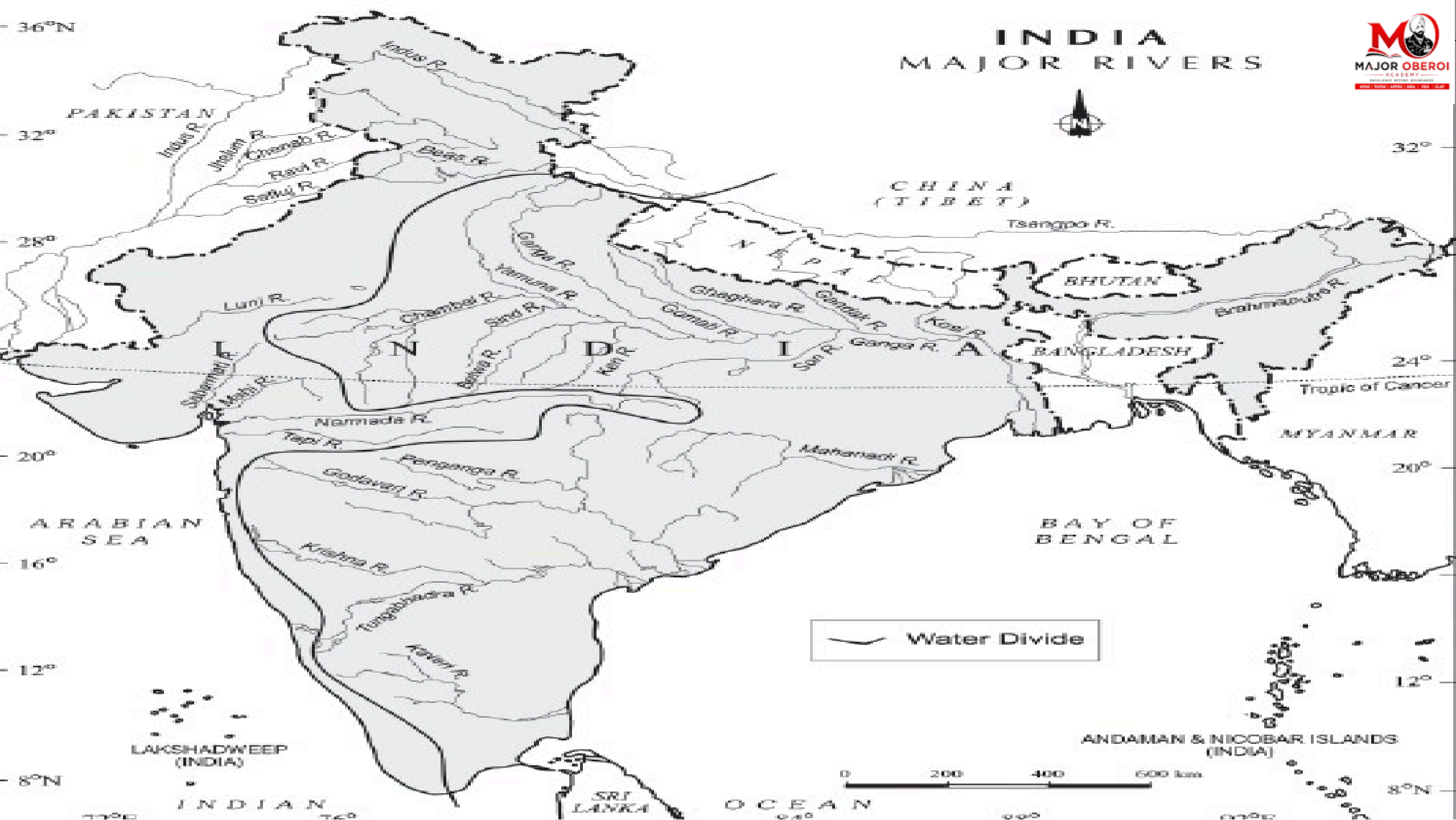
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- Classification of Drainage Systems in India:
- 1. On the basis of discharge of water – the Arabian Sea drainage and the Bay of Bengal drainage
- On the basis of discharge of water (orientations to the sea), the drainage system of India may be grouped into:
- the Arabian Sea drainage
- the Bay of Bengal drainage

- These two drainage systems are separated from each other by the Delhi ridge, the Aravalis and the Sahyadris.
- Nearly 77 percent of the drainage area consisting of the Ganga, the Brahmaputra, the Mahanadi, the Godavari, the Krishna, etc. is oriented towards the Bay of Bengal while 23 percent comprising the Indus, the Narmada, the Tapti, the Mahi and the Periyar systems discharge their waters into the Arabian Sea.

- 1. Rivers of the inland drainage basin (endorheic basin): When a river does not reach the sea but disappears into the sand, such a region is called an area of inland drainage. Inland drainage streams are ephemeral streams (short-lived).
- E.g The Ghaggar river in Haryana, which is supposed to be remnant of the proverbial ancient Saraswati river, gets lost in dry sands near Hanumangarh in Rajasthan.

- The Luni river in Rajasthan originates near Ajmer and after flowing through Thar desert gets lost in the Rann of Kutch.
- The Aksai Chin region in Ladakh too has some inland drainage.


INDIA
MAJOR RIVERS
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PAKISTAN
CHINA (TIBET)
NEPAL
BHUTAN
BANGLADESH
MYANMAR
ARABIAN SEA
BAY OF BENGAL
INDIAN OCEAN
SRI LANKA
LAKSHADWEEP (INDIA)
ANDAMAN & NICOBAR ISLANDS (INDIA)
Indus R.
Jhelum R.
Chenab R.
Ravi R.
Sutlej R.
Beas R.
Ganga R.
Yamuna R.
Ghaghara R.
Gandak R.
Kosi R.
Brahmaputra R.
Tsangpo R.
Luni R.
Chambal R.
Sind R.
Betwa R.
Ken R.
Son R.
Sabarmati R.
Mahi R.
Narmada R.
Tapi R.
Mahanadi R.
Godavari R.
Penganga R.
Krishna R.
Tungabhadra R.
Kaveri R.
Tropic of Cancer
Water Divide
0 200 400 600 km
36°N
32°
28°
24°
20°
16°
12°
8°N

- On the basis of the size of the watershed:
- Major river basins – with more than 20,000 sq. km of catchment area. It includes 14 drainage basins such as the Ganga, the Brahmaputra, the Godavari, the Krishna, the Tapti, the Narmada, the Mahi, the Pennar, the Sabarmati, the Barak, etc.

- Medium river basins – with catchment area between 2,000-20,000 sq. km incorporating 44 river basins such as the Kalindi, the Periyar, the Meghna, etc.
- Minor river basins – with catchment area of less than 2,000 sq. Km, these include 55 river basins.

- 3.On the basis of the mode of origin, nature and characteristics:
- The Himalayan drainage, and
- The Peninsular drainage.
- There is no clear-cut line of demarcation between these two drainage systems, as many of the peninsular rivers like the Chambal, Betwa, Sind, Ken and Son are much older in age and origin than the Himalayan rivers.

- **The Himalayan Drainage system**:
- It consists of the rivers originating in the Himalayan and trans-Himalayan region. It further consists of three river systems namely:
- the Ganga,
- the Indus, and
- the Brahmaputra river systems.

- Since these are fed both by the melting of snow and precipitation, rivers of this system are perennial.
- The various geographical features made by the Himalayan rivers are:
- In upper reaches (Youthful stage): Gorges, V-shaped valleys, rapids, waterfalls, truncated spurs etc.

- In plain areas or middle part (Mature stage): While entering the plains, they form depositional features like flat valleys, ox-bow lakes, flood plains, braided channels, and deltas near the river mouth. Over the plains, they display a strong meandering tendency and shift their courses frequently.

Evolution of the Himalayan Drainage System:

- There is a difference of opinion about the evolution of the Himalayan Rivers. However, geologists believe that:
- A mighty river called Shiwalik or Indo-Brahma traversed the entire longitudinal extent of the Himalaya from Assam to Punjab and onwards to Sind, and finally discharged into the Gulf of Sind near lower Punjab during the Miocene period some 5-24 million years ago.

- The remarkable continuity of the Shiwalik and its lacustrine origin and alluvial deposits consisting of sands, silt, clay, boulders and conglomerates support this viewpoint.

- It is opined that in due course of time Indo–Brahma river was dismembered into three main drainage systems:
- the Indus and its five tributaries in the western part;
- the Ganga and its Himalayan tributaries in the central part; and
- the stretch of the Brahmaputra in Assam and its Himalayan tributaries in the eastern part.

- The dismemberment was probably due to the Pleistocene upheaval in the western Himalayas, including the uplift of the Potwar Plateau (Delhi Ridge), which acted as the water divide between the Indus and Ganga drainage systems.

- Likewise, the downthrusting of the Malda gap area between the Rajmahal hills and the Meghalaya plateau during the mid-pleistocene period, diverted the Ganga and the Brahmaputra systems to flow towards the Bay of Bengal.

- **The Ganga River System**
- The Ganga is the most important river of India both from the point of view of its basin and cultural significance. It rises from the Gangotri glacier near Gaumukh (3,900 m) in the Uttarkashi district of Uttarakhand. Here, it is known as the Bhagirathi. The important tributaries of Ganga are the Ramganga, the Gomati, the Ghaghara, the Gandak, the Kosi and the Mahanada. The river finally discharges itself into the Bay of Bengal near the Sagar Island.

- It rises from the Gangotri glacier near Gaumukh (3,900 m) in the Uttarkashi district of Uttarakhand.
- Here, it is known as the Bhagirathi.
- It cuts through the Central and the Lesser Himalayas in narrow gorges.

- At Devprayag, the Bhagirathi meets the Alaknanda; then after, it is known as the Ganga. The Alaknanda has its source in the Satopanth glacier above Badrinath. The Alaknanda consists of the Dhauli and the Vishnu Ganga which meet at Joshimath or Vishnu Prayag. The other tributaries of Alaknanda such as the Pindar join it at Karna Prayag while Mandakini or Kali Ganga meets it at Rudra Prayag.
- The Ganga enters the plains at Haridwar.

Place	Confluence
Devprayag	Bhagirathi + Alaknanda
Rudraprayag	Mandakini + Alaknanda
Karnaprayag	Pindar + Alaknanda
Vishnuprayag	Dhauliganga + Alaknanda

- From Haridwar, it flows first to the south, then to the south-east and east before splitting into two tributaries, namely the Bhagirathi and the Hugli. The river has a length of 2,525 km. It is shared by Uttarakhand (110 km) and Uttar Pradesh (1,450 km), Bihar (445 km) and West Bengal (520 km). The Ganga basin covers about 8.6 lakh sq. km area in India alone. The Ganga river system is the largest in India having a number of perennial and non-perennial rivers originating in the Himalayas in the north and the Peninsula in the south, respectively.

- The important tributaries of Ganga are the Ramganga, the Gomati, the Ghaghara, the Gandak, the Kosi and the Mahanada. The river finally discharges itself into the Bay of Bengal near the Sagar Island.

The Brahmaputra River System

- The river Brahmaputra originates in Himalayan Lake Manasarover in Tibet to the outfall in the Bay of Bengal. It flows eastward in Tibet and south, south-west in India and traverses a distance of about 2900 km out of which 1,700km is in Tibet, 900 km is in India and 300 km is in Bangladesh. It is fed by the glaciers and joined by a number of tributaries which originates at different elevation in the hills encircling the catchment forming watershed.


CHINA
New Delhi
NEPAL
Lucknow
Kathmandu
Jaipur
Agra
Kanpur
Ganges
Allahabad
Patna
Farakka
Thimphu
Kolkata
BHUTAN
Brahmaputra
Dhaka
Sylhet
INDIA
MYANMAR
Bay of Bengal

- The Brahmaputra River System (3848 km) is one of the longest rivers of the world. It is known as the Yarlung Tsangpo River in Tibet, the Brahmaputra, Lohit, Siang, and Dihang in India, and the Jamuna in Bangladesh.

- It traverses eastward longitudinally for a distance of nearly 1,200 km in a dry and flat region of southern Tibet, where it is known as the Tsangpo, which means ‘the purifier.’ The Rango Tsangpo is the major right bank tributary of this river in Tibet. It emerges as a turbulent and dynamic river after carving out a deep gorge in the Central Himalayas near Namcha Barwa (7,755 m).

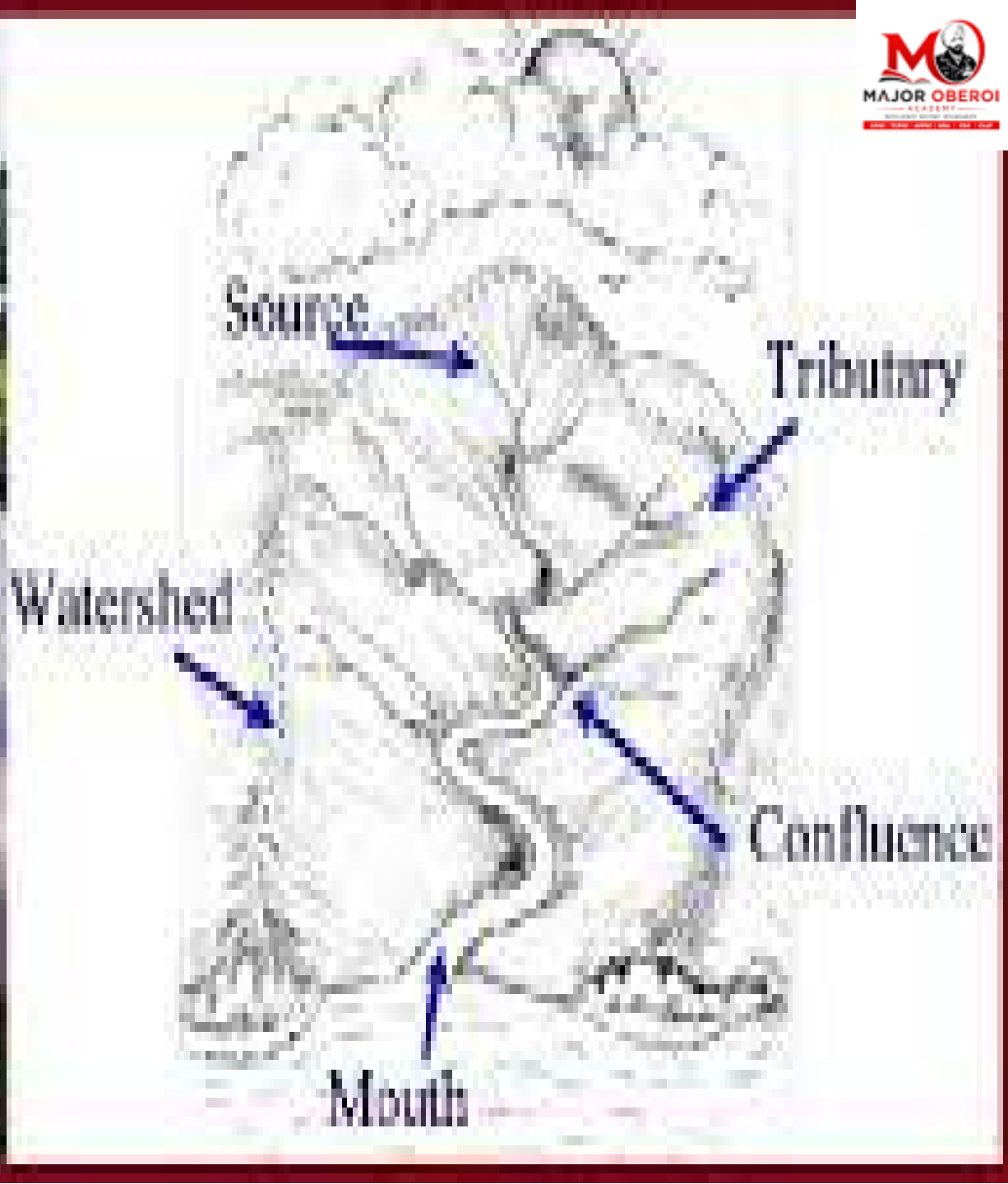
- The river emerges from the foothills under the name of Siang or Dihang. It enters India west of Sadiya town in Arunachal Pradesh.
- Flowing southwest, it receives its main left bank tributaries, viz., Dibang or Sikang and Lohit; thereafter, it is known as the Brahmaputra. Among the tributaries Subansiri, Manas, Jiabharali, Pagladiya, Puthimari and Sankosh etc. are snow fed.

- It receives numerous tributaries in its 750 km long journey through the Assam valley. Its major left bank tributaries are the Burhi Dihing, Dhansari (South) and Kalang whereas the important right bank tributaries are the Subansiri, Kameng, Manas and Sankosh. The Subansiri which has its origin in Tibet is an antecedent river.

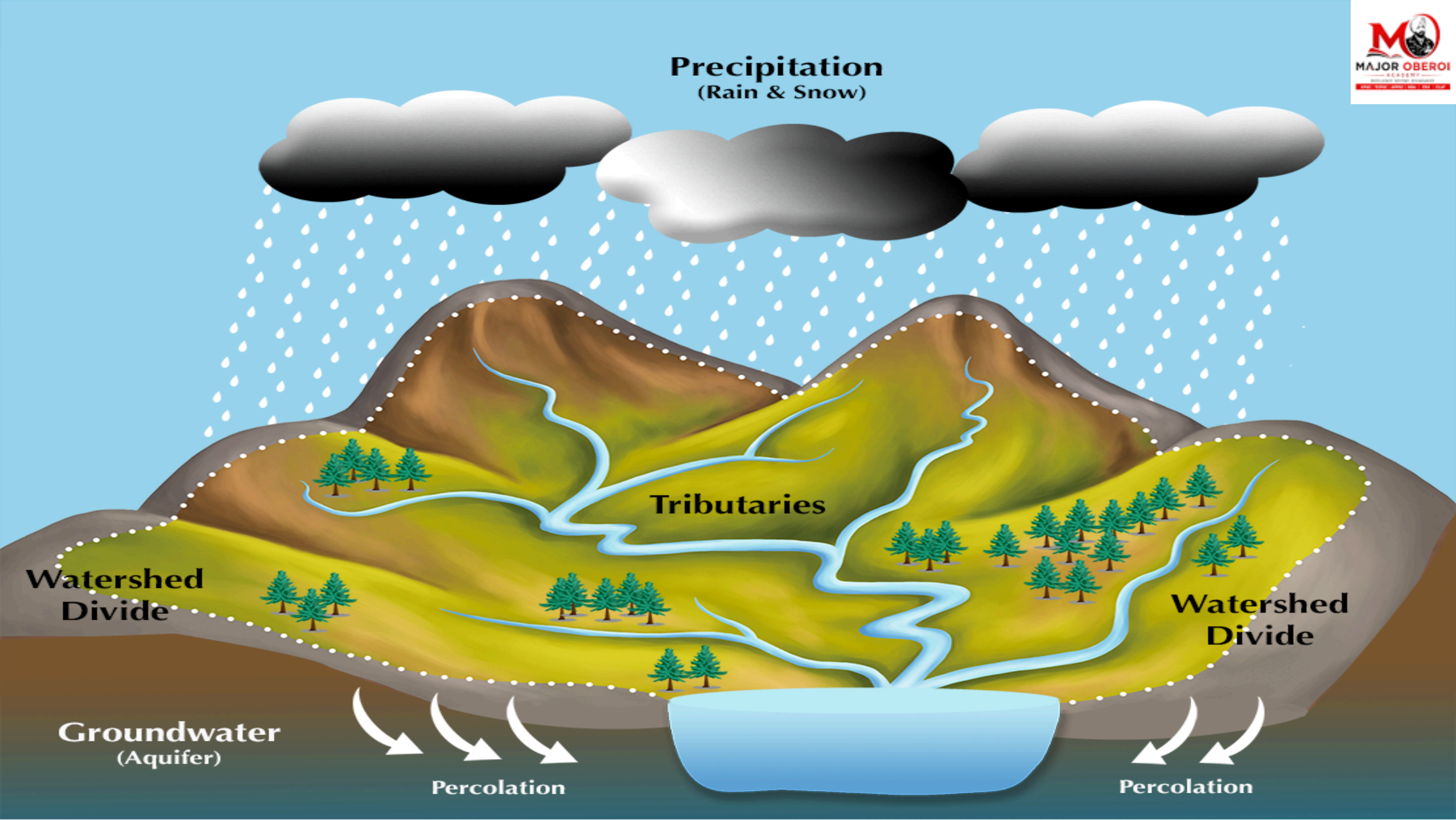
What is an Antecedent River?

- An antecedent stream is a stream that maintains its original course and pattern despite the changes in underlying rock topography. A stream with a dendritic drainage pattern, for example, can be subject to slow tectonic uplift. ... The stream erodes a gorge in the resistant bed and continues its flow as before.




Source
Tributary
Watershed
Confluence
Mouth

- A watershed is an area of land that drains or “sheds” water into a specific waterbody. Every body of water has a watershed. Watersheds drain rainfall and snowmelt into streams and rivers. These smaller bodies of water flow into larger ones, including lakes, bays, and oceans.


Precipitation
(Rain & Snow)
Tributaries
Watershed
Divide
Watershed
Divide
Groundwater
(Aquifer)
Percolation
Percolation

- The Brahmaputra enters into Bangladesh near Dhubri and flows southward. In Bangladesh, the Tista joins it on its right bank from where the river is known as the Jamuna. It finally merges with the river Padma, which falls in the Bay of Bengal. The Brahmaputra is well-known for floods, channel shifting and bank erosion. This is due to the fact that most of its tributaries are large, and bring large quantity of sediments owing to heavy rainfall in its catchment area.